MLS_Chap2_p16

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1 MLS Chapter 2 problem 16 p. 77

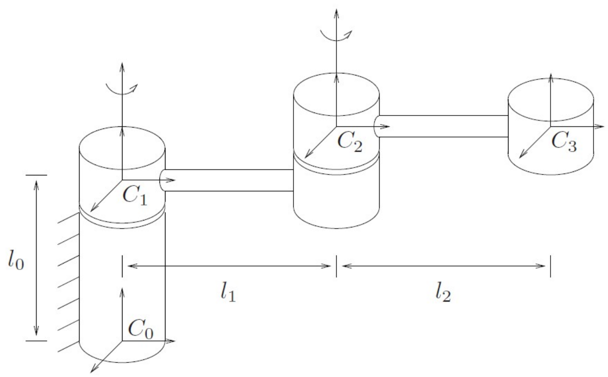

Figure 2.17: A two degree of freedom manipulator.

By choice, the problem is solved here with ξ_2 defined in frame C_1 (because q_2 will be defined in frame C_1).

[3]:
```
from sympy import symbols, simplify, diff, Function
from sympy.matrices import Matrix, BlockMatrix, zeros, eye
from robotics_functions_sym import (hat, Rodrigues, wedge, vee, gmat,
                  gmati,Adgg,Adggi)
```

[4]:
```
t, l_0, l_1, l_2     = symbols('t, l_0, l_1, l_2',real=True)
theta1  = Function('theta1')(t)
theta1
```

[4]: $\theta_1(t)$

[5]:
```
theta2  = Function('theta2')(t)
theta2
```

[5]: $\theta_2(t)$

[6]:
```
u = Matrix([[0],[0],[1]])
exphutheta1 = Rodrigues(u,theta1)  # creating rotation matrix with Rodrigues'
     # formula, MLS eq. (2.14) p. 28; u is the direction unit vector
exphutheta2 = Rodrigues(u,theta2)
```

The next three cells calculate the initial configuratons of frames C_1, C_2 and C_3, $g_{c0c1}(0)$, $g_{c1c2}(0)$ and $g_{c2c3}(0)$, respectively, with the function `gmat`. In the initial configurations, $\theta_1 = \theta_2 = 0$ and hence the rotation matrices are all identity matrices.

[8]:
```
p_c0c10 = Matrix([[0],[0],[l_0]])
gc0c10  = gmat(eye(3),p_c0c10)
gc0c10
```

[8]:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[9]:
```
p_c1c20  = Matrix([[0],[l_1],[0]])  # I assume here that the frame $C2$ origin
                                    #  has the same $z$ coordinate as frame $C1$.
gc1c20  = gmat(eye(3),p_c1c20)
gc1c20
```

[9]:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[10]:
```
p_c2c30  = Matrix([[0],[l_2],[0]])  # I assume here that the frame $C3$ origin
                                    #  has the same $z$ coordinate as frame $C2$.
gc2c30  = gmat(eye(3),p_c2c30)
gc2c30
```

[10]:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

From eq. (2.45) p. 49:

$$g_{c0c1}(\theta_1) = e^{\hat{\xi}_1 \theta_1} g_{c0c1}(0)$$

To calculate $e^{\hat{\xi}_1 \theta_1}$ I shall use MLS eq. (2.36) p. 42. On p. 47 MLS states (and this has been proven in the slides set for March 16, 2026) that if we choose $v = -\hat{\omega} q + h\omega$, the twist $\xi = (v, \omega)$ defined in eq. (2.36) generates the screw defined in eq. (2.40). For a revolute joint, like the two joints in this example, $h = 0$ and it was shown in the notes set "on_MLS_Chapter_2_Section_2_3.pdf" (on Moodle) that, when $h = 0$, the term $\omega\omega^T v\theta = 0$. Therefore, for this example

$$v_1 = -\hat{\omega}_1 q_1$$

and

$$e^{\hat{\xi}_1 \theta_1} = \begin{bmatrix} e^{\hat{\omega}_1 \theta_1} & (I - e^{\hat{\omega}_1 \theta_1})\hat{\omega}_1 v_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^{\hat{\omega}_1 \theta_1} & (I - e^{\hat{\omega}_1 \theta_1}) q_1 \\ 0 & 1 \end{bmatrix},$$

by equations (2.36) and (2.40). Note that in this case, $\omega_1 = \omega_2$ and is equal to the unit vector `u` in the code.

Similarly for $g_{c1c2}(\theta_2)$.

q_1 is a point (any point) on the axis of ξ_1 (expressed in frame C_0), q_2 is a point (any point) on the axis of ξ_2 (expressed in frame C_1).

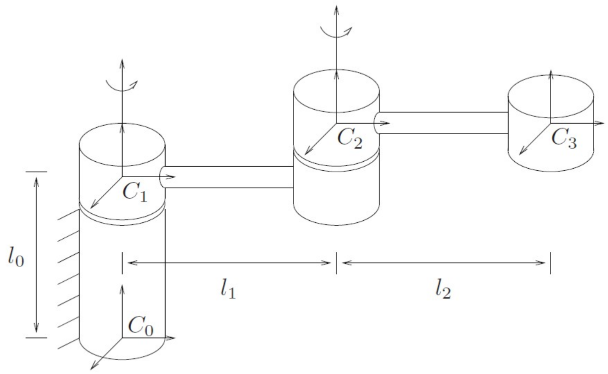

Figure 2.17: A two degree of freedom manipulator.

[12]:
```
q1 = Matrix([[0],[0],[0]])
q2 = Matrix([[0],[l_1],[0]])
q3 = Matrix([[0],[l_2],[0]])
```

[13]:
```
v1 = -hat(u)*q1
v1
```

[13]:

$$\begin{bmatrix}0\\0\\0\end{bmatrix}$$

[14]:
```
v2 = -hat(u)*q2
v2
```

[14]:

$$\begin{bmatrix}l_1\\0\\0\end{bmatrix}$$

[15]:
```
v3 = -hat(u)*q3
v3
```

[15]:

$$\begin{bmatrix}l_2\\0\\0\end{bmatrix}$$

[16]:
```
twistexp1 =  Matrix(BlockMatrix(
    [[exphutheta1,(eye(3) - exphutheta1)*q1],[zeros(1,3),Matrix([1])]]
                                ))
twistexp1
```

[16]:

$$\begin{bmatrix}\cos(\theta_1(t)) & -\sin(\theta_1(t)) & 0 & 0\\ \sin(\theta_1(t)) & \cos(\theta_1(t)) & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

[17]:
```
twistexp2 =  Matrix(BlockMatrix(
    [[exphutheta2,(eye(3) - exphutheta2)*q2],[zeros(1,3),Matrix([1])]]
                                ))
twistexp2
```

[17]:

$$\begin{bmatrix}\cos(\theta_2(t)) & -\sin(\theta_2(t)) & 0 & l_1\sin(\theta_2(t))\\ \sin(\theta_2(t)) & \cos(\theta_2(t)) & 0 & l_1\cdot(1-\cos(\theta_2(t)))\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

[18]:
```
twistexp3 =  Matrix(BlockMatrix(
    [[eye(3),(eye(3) - eye(3))*q3],[zeros(1,3),Matrix([1])]]
                                ))
twistexp3
```

[18]:

$$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

[19]:
```
gc0c1 = twistexp1*gc0c10
gc0c1
```

[19]:

$$\begin{bmatrix}\cos(\theta_1(t)) & -\sin(\theta_1(t)) & 0 & 0\\ \sin(\theta_1(t)) & \cos(\theta_1(t)) & 0 & 0\\ 0 & 0 & 1 & l_0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

[20]:
```
gc1c2 = simplify(twistexp2*gc1c20)
gc1c2
```

[20]:

$$\begin{bmatrix}\cos(\theta_2(t)) & -\sin(\theta_2(t)) & 0 & 0\\ \sin(\theta_2(t)) & \cos(\theta_2(t)) & 0 & l_1\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

[21]:
```
gc2c3 = simplify(twistexp3*gc2c30)
gc2c3
```

[21]:

$$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & l_2\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

1.1 Part (a)

Using eq. (2.24) p. 37:

$$g_{c0c3} = g_{c0c1}g_{c1c2}g_{c2c3}\,.$$

[23]:
```
gc0c3 = simplify(gc0c1*gc1c2*gc2c3)
gc0c3
```

[23]:

$$\left[\begin{matrix}\cos\left(\theta_{1}(t)+\theta_{2}(t)\right) & -\sin\left(\theta_{1}(t)+\theta_{2}(t)\right) & 0 & -l_{1}\sin\left(\theta_{1}(t)\right)-l_{2}\sin\left(\theta_{1}(t)+\theta_{2}(t)\right)\\ \sin\left(\theta_{1}(t)+\theta_{2}(t)\right) & \cos\left(\theta_{1}(t)+\theta_{2}(t)\right) & 0 & l_{1}\cos\left(\theta_{1}(t)\right)+l_{2}\cos\left(\theta_{1}(t)+\theta_{2}(t)\right)\\ 0 & 0 & 1 & l_{0}\\ 0 & 0 & 0 & 1\end{matrix}\right]$$

R_{c0c3} :

[25]:
```
Rc0c3 = gc0c3[:3,:3]
Rc0c3
```

[25]:

$$\left[\begin{matrix}\cos\left(\theta_{1}(t)+\theta_{2}(t)\right) & -\sin\left(\theta_{1}(t)+\theta_{2}(t)\right) & 0\\ \sin\left(\theta_{1}(t)+\theta_{2}(t)\right) & \cos\left(\theta_{1}(t)+\theta_{2}(t)\right) & 0\\ 0 & 0 & 1\end{matrix}\right]$$

p_{c0c3} :

[27]:
```
pc0c3 = gc0c3[:3,3]
pc0c3
```

[27]:

$$\left[\begin{matrix}-l_{1}\sin\left(\theta_{1}(t)\right)-l_{2}\sin\left(\theta_{1}(t)+\theta_{2}(t)\right)\\ l_{1}\cos\left(\theta_{1}(t)\right)+l_{2}\cos\left(\theta_{1}(t)+\theta_{2}(t)\right)\\ l_{0}\end{matrix}\right]$$

1.2 Part (b)

Calculation of V^s_{c0c1}

MLS p. 54: $\hat{V}^s_{c0c1} = \dot{g}_{c0c1} g^{-1}_{c0c1}$ and MLS p. 58: $\hat{V}^s_{c0c1} = \hat{\xi}_1 \dot{\theta}_1$:

[29]:
```
Vhatc0c1s = simplify(diff(gc0c1,t)*gmati(gc0c1))
Vhatc0c1s
```

[29]:

$$\left[\begin{matrix}0 & -\frac{d}{dt}\theta_{1}(t) & 0 & 0\\ \frac{d}{dt}\theta_{1}(t) & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{matrix}\right]$$

[30]:
```
hat_xi1      = wedge(Matrix(BlockMatrix([[v1],[u]])))
Vhatc0c1s_2 = hat_xi1*diff(theta1,t)

shouldbezero4times4 = Vhatc0c1s - Vhatc0c1s_2
shouldbezero4times4
```

[30]:

$$\left[\begin{matrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{matrix}\right]$$

[31]:
```
Vc0c1s = vee(Vhatc0c1s)  # vee operator function, like in MLS eq. (2.30)
                         #  p. 41
Vc0c1s
```

[31]:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) \end{bmatrix}$$

Calculation of V^s_{c1c2}

MLS p. 54: $\hat{V}^s_{c1c2} = \dot{g}_{c1c2} g^{-1}_{c1c2}$ and MLS p. 58: $\hat{V}^s_{c1c2} = \hat{\xi}_2 \dot{\theta}_2$:

[33]:
```
Vhatc1c2s = simplify(diff(gc1c2,t)*gmati(gc1c2))
Vhatc1c2s
```

[33]:

$$\begin{bmatrix} 0 & -\frac{d}{dt}\theta_2(t) & 0 & l_1 \frac{d}{dt}\theta_2(t) \\ \frac{d}{dt}\theta_2(t) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[34]:
```
hat_xi2     = wedge(Matrix(BlockMatrix([[v2],[u]])))
Vhatc1c2s_2 = hat_xi2*diff(theta2,t)
```

[35]:
```
shouldbezero4times4 = Vhatc1c2s - Vhatc1c2s_2
shouldbezero4times4
```

[35]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[36]:
```
Vc1c2s = vee(Vhatc1c2s)  # vee operator function, like in MLS eq. (2.30)
                     #  p. 41
Vc1c2s
```

[36]:

$$\begin{bmatrix} l_1 \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

Calculation of V^s_{c2c3}

MLS p. 54: $\hat{V}^s_{c2c3} = \dot{g}_{c2c3} g^{-1}_{c2c3}$ and MLS p. 58: $\hat{V}^s_{c2c3} = \hat{\xi}_3 \dot{\theta}_3 = 0$ as $\dot{\theta}_3 = 0$ and $\dot{g}_{c2c3} = 0$:

[38]:
```
Vhatc2c3s = simplify(diff(gc2c3,t)*gmati(gc2c3))
Vhatc2c3s
```

[38]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[39]:
```
Vc2c3s = vee(Vhatc2c3s)  # vee operator function, like in MLS eq. (2.30)
                         #  p. 41
Vc2c3s
```

[39]:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Using Proposition 2.14 p. 58:

$$V^s_{c1c3} = V^s_{c1c2} + Ad_{g_{c1c2}} V^s_{c2c3} = V^s_{c1c2}$$

since $V^s_{c2c3} = 0$.

[41]:
```
Vc1c3s = Vc1c2s
Vc1c3s
```

[41]:

$$\begin{bmatrix} l_1 \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

Using Proposition 2.14 p. 58:

$$V^s_{c0c3} = V^s_{c0c1} + Ad_{g_{c0c1}} V^s_{c1c3} \,.$$

[43]:
```
Ad_gc0c1 = Adgg(gc0c1)  # calculation of adjoint transformation matrix,
                        # MLS p. 55
Ad_gc0c1
```

[43]:

$$\begin{bmatrix} \cos(\theta_1(t)) & -\sin(\theta_1(t)) & 0 & -l_0 \sin(\theta_1(t)) & -l_0 \cos(\theta_1(t)) & 0 \\ \sin(\theta_1(t)) & \cos(\theta_1(t)) & 0 & l_0 \cos(\theta_1(t)) & -l_0 \sin(\theta_1(t)) & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta_1(t)) & -\sin(\theta_1(t)) & 0 \\ 0 & 0 & 0 & \sin(\theta_1(t)) & \cos(\theta_1(t)) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

[44]:
```
Vc0c3s = Vc0c1s + Ad_gc0c1*Vc1c3s
Vc0c3s
```

[44]:

$$\begin{bmatrix} l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

1.2.1 Alternatively

$$\hat{V}^s_{c0c3} = \dot{g}_{c0c3} g^{-1}_{c0c3} \ .$$

[46]:
```
Vhatc0c3s_alt = simplify(diff(gc0c3,t)*gmati(gc0c3))
Vhatc0c3s_alt
```

[46]:
$$\begin{bmatrix} 0 & -\frac{d}{dt}\theta_1(t) - \frac{d}{dt}\theta_2(t) & 0 & l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) & 0 & 0 & l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[47]:
```
Vc0c3s_alt = vee(Vhatc0c3s_alt)
Vc0c3s_alt
```

[47]:
$$\begin{bmatrix} l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

[48]:
```
shouldbezero6times1 = Vc0c3s - Vc0c3s_alt
shouldbezero6times1
```

[48]:
$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

1.3 Part (c)

By proposition 2.15 p. 59:

$$V^b_{c0c3} = \mathrm{Ad}_{g^{-1}_{c2c3}} V^b_{c0c2} + V^b_{c2c3} = \mathrm{Ad}_{g^{-1}_{c2c3}} V^b_{c0c2} \ ,$$

since $V^b_{c2c3} = 0$.

By eq. (2.55) p. 55:

$$\hat{V}^b_{c0c2} = g^{-1}_{c0c2} \dot{g}_{c0c2} \ ,$$

where

$$g_{c0c2} = g_{c0c1} g_{c1c2} \ ,$$

using eq. (2.24) p. 37.

[51]:
```
gc0c2      = gc0c1*gc1c2
Vhatcoc2b = simplify(gmati(gc0c2)*diff(gc0c2,t))
Vcoc2b     = vee(Vhatcoc2b)
Vcoc2b
```

[51]:

$$\begin{bmatrix} -l_1 \cos\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) \\ l_1 \sin\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

[52]:
```
Vcoc3b = Adggi(gc2c3)*Vcoc2b
Vcoc3b
```

[52]:

$$\begin{bmatrix} -l_1 \cos\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) - l_2 \left(\frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t)\right) \\ l_1 \sin\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

1.3.1 Alternative method

using the eq. (2.61) p. 56 and the result of part (b):

$$V^s_{c0c3} = \mathrm{Ad}_{g_{c0c3}} V^b_{c0c3}\ .$$

Therefore,

$$V^b_{c0c3} = \mathrm{Ad}_{g^{-1}_{c0c3}} V^s_{c0c3}\ .$$

[54]:
```
Vcoc3b_alt = simplify(Adggi(gc0c3)*Vc0c3s)
Vcoc3b_alt
```

[54]:

$$\begin{bmatrix} -l_1 \cos\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) - l_2 \frac{d}{dt}\theta_1(t) - l_2 \frac{d}{dt}\theta_2(t) \\ l_1 \sin\left(\theta_2(t)\right) \frac{d}{dt}\theta_1(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

[55]:
```
shouldbezero6times1 = simplify(Vcoc3b - Vcoc3b_alt)
shouldbezero6times1
```

[55]:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

1.3.2 Yet another alternative method

using the eq. (2.55) p. 55:

$$\hat{V}^b_{c0c3} = g^{-1}_{c0c3}\dot{g}_{c0c3} \; .$$

[57]:
```
Vhatcoc3b_alt2 = simplify(gmati(gc0c3)*diff(gc0c3,t))
Vhatcoc3b_alt2
```

[57]:
$$\begin{bmatrix} 0 & -\frac{d}{dt}\theta_1(t) - \frac{d}{dt}\theta_2(t) & 0 & -l_1\cos(\theta_2(t))\frac{d}{dt}\theta_1(t) - l_2\frac{d}{dt}\theta_1(t) - l_2\frac{d}{dt}\theta_2(t) \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) & 0 & 0 & l_1\sin(\theta_2(t))\frac{d}{dt}\theta_1(t) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

[58]:
```
Vcoc3b_alt2          = vee(Vhatcoc3b_alt2)
shouldbezero6times1 = simplify(Vcoc3b - Vcoc3b_alt2)
shouldbezero6times1
```

[58]:
$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

1.4 Part (d)

Let q_{c3} be the coordinates in homogeneous form of the frame C_3 origin, expressed in frame C_2. Then, using the first of eq. (2.56) p. 55, the velocity of the frame C_3 origin expressed in frame C_2 is

$$v_{q_{c3_b}} = \hat{V}^b_{c0c2}q_{c3} \; .$$

$\hat{V}^b_{c0c2}$ has already been calculated in part (c) above.

[60]:
```
qc3 = Matrix([[0],[l_2],[0],[1]])
qc3
```

[60]:
$$\begin{bmatrix} 0 \\ l_2 \\ 0 \\ 1 \end{bmatrix}$$

[61]:
```
vqc3b = simplify(Vhatcoc2b*qc3)
vqc3b
```

[61]:
$$\begin{bmatrix} -l_1 \cos(\theta_2(t)) \frac{d}{dt}\theta_1(t) - l_2 \frac{d}{dt}\theta_1(t) - l_2 \frac{d}{dt}\theta_2(t) \\ l_1 \sin(\theta_2(t)) \frac{d}{dt}\theta_1(t) \\ 0 \\ 0 \end{bmatrix}$$

[62]:
```
shouldbezero3times1 = simplify(Vcoc3b[:3,0] - vqc3b[:3,0])
shouldbezero3times1
```

[62]:
$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let q_{c3}^{c0} be the coordinates in homogeneous form of the frame C_3 origin expressed in frame C_0. Then, using the first of eq. (2.54) p. 54, the velocity of the origin of frame C_3 expressed in frame C_0 is

$$v_{q_{c3_s}} = v_{q_{c3}^{c0}} = \hat{V}_{c0c2}^s q_{c3}^{c0} \ .$$

Using eq. (2.24) p. 37:

$$g_{c0c2} = g_{c0c1} g_{c1c2} \ .$$

Using eq. (2.22) p. 35, expressed in homogeneous form:

$$q_{c3}^{c0} = g_{c0c2} q_{c3} \ .$$

[64]:
```
gc0c2 = simplify(gc0c1*gc1c2)
gc0c2
```

[64]:
$$\begin{bmatrix} \cos(\theta_1(t) + \theta_2(t)) & -\sin(\theta_1(t) + \theta_2(t)) & 0 & -l_1 \sin(\theta_1(t)) \\ \sin(\theta_1(t) + \theta_2(t)) & \cos(\theta_1(t) + \theta_2(t)) & 0 & l_1 \cos(\theta_1(t)) \\ 0 & 0 & 1 & l_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[65]:
```
qc3_c0 = gc0c2*qc3
qc3_c0
```

[65]:
$$\begin{bmatrix} -l_1 \sin(\theta_1(t)) - l_2 \sin(\theta_1(t) + \theta_2(t)) \\ l_1 \cos(\theta_1(t)) + l_2 \cos(\theta_1(t) + \theta_2(t)) \\ l_0 \\ 1 \end{bmatrix}$$

By eq. (2.61) p. 56:

$$V_{c0c2}^s = \mathrm{Ad}_{g_{c0c2}} V_{c0c2}^b \ .$$

V_{c0c2}^b has already been calculated in part (c) above.

[67]:
```
Ad_gc0c2 = Adgg(gc0c2)  # calculation of adjoint transformation matrix,
                        # MLS p. 55
Vcoc2s   = simplify(Ad_gc0c2*Vcoc2b)
Vcoc2s
```

[67]:

$$\begin{bmatrix} l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \\ 0 \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \end{bmatrix}$$

[68]:
```
simplify(simplify(diff(gc0c2,t)*gmati(gc0c2)) - wedge(Vcoc2s))
```

[68]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

As stated above, where q_{c3}^{c0} is the coordinates in homogeneous form of frame C_3 expressed in frame C_0, using the first of eq. (2.54) p. 54, the spatial velocity of the origin of frame C_3 is

$$v_{q_{c3_s}} = v_{q_{c3}^{c0}} = \hat{V}_{c0c2}^{s} q_{c3}^{c0} \,.$$

[70]:
```
vqc3s = simplify(wedge(Vcoc2s)*qc3_c0)
vqc3s
```

[70]:

$$\begin{bmatrix} -l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_1(t) - l_2 \cos(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_1(t) - l_2 \cos(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_2(t) \\ -l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_1(t) - l_2 \sin(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_1(t) - l_2 \sin(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \end{bmatrix}$$

[71]:
```
shouldbezero4times1 = vqc3s - simplify(gc0c2*vqc3b)
shouldbezero4times1
```

[71]:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Above we calculated $v_{q_{c3_s}} = v_{q_{c3}^{c0}}$ using

$$v_{q_{c3_s}} = v_{q_{c3}^{c0}} = \hat{V}_{c0c2}^{s} q_{c3}^{c0} \,.$$

But we can also calculate $v_{q_{c3_s}} = v_{q_{c3}^{c0}}$, also using the first of eq. (2.54) p. 54, as

$$v_{q_{c3_s}} = v_{q_{c3}^{c0}} = \hat{V}_{c0c3}^{s} q_{c3}^{c0} \,.$$

This can be true only if $V_{c0c3}^{s} = V_{c0c2}^{s}$.

By proposition 2.14 p. 58:

$$V^s_{c0c3} = V^s_{c0c2} + \mathrm{Ad}_{g_{c0c2}} V^s_{c2c3} = V^s_{c0c2} + \mathrm{Ad}_{g_{c0c2}} V^{c2}_{c2c3} \ .$$

It was shown above, in part (a), that $V^s_{c2c3} = 0$. Therefore, indeed, $V^s_{c0c3} = V^s_{c0c2}$.

[73]:
```
qc3_c3     = Matrix([[0],[0],[0],[1]])  # Coordinate of the origin of frame C3␣
 ↪expressed in frame C3
qc3_c0_alt = gc0c3*qc3_c3
dum        = simplify(Vhatc0c3s_alt*qc3_c0_alt)  # 1st of eq. (2.54) p. 54
dum
```

[73]:

$$\begin{bmatrix} -l_1 \cos(\theta_1(t)) \frac{d}{dt}\theta_1(t) - l_2 \cos(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_1(t) - l_2 \cos(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_2(t) \\ -l_1 \sin(\theta_1(t)) \frac{d}{dt}\theta_1(t) - l_2 \sin(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_1(t) - l_2 \sin(\theta_1(t) + \theta_2(t)) \frac{d}{dt}\theta_2(t) \\ 0 \\ 0 \end{bmatrix}$$

[74]:
```
shoudlbezero4times1 = vqc3s - dum
shoudlbezero4times1
```

[74]:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$